

1. **MIMD stands for.....**
 - A. Multiple instruction multiple data
 - B. Memory instruction multiple data
 - C. Multiple instruction memory data
 - D. Multiple information memory data
2. **Logic gates with a set of input and an output is arrangement of....**
 - A. Computational circuit
 - B. Design circuit
 - C. Logic circuit
 - D. Register
3. **Which one of the following is not a valid rule of Boolean algebra?**
 - A. $A + 1 = 1$
 - B. $A.A = A$
 - C. $A = \bar{A}$
 - D. $A + 0 = A$
4. **The Boolean expression $AB'CD'$ is:**
 - A. Sum term
 - B. A literal term
 - C. Product term
 - D. Always 1
5. **If two adjacent 1s are detected in the input, the output is set to high. Input combinations will be:**
 - A. 0011
 - B. 1100
 - C. 0101
 - D. 1010
6. **Flip flops are also called....**
 - A. Bi-stable dual vibrators
 - B. Bi-stable multi vibrators
 - C. Bi-stable transformer
 - D. Bi-stable single vibrators
7. **.... occurs when the same clock signal arrives at different times at different clock inputs due to propagation delay.**
 - A. Race condition
 - B. Ripple effect
 - C. Clock skew
 - D. None of given options
8. **The most important feature of spiral model is:**
 - A. Requirement analysis
 - B. Quality management
 - C. Risk management
 - D. Configuration management
9. **All the modules of the system are integrated and tested as complete system in the case of:**
 - A. Bottom up testing
 - B. Sandwich testing
 - C. Top-down testing
 - D. Big-Bang testing
10. **Which of the following statement is true?**
 - A. Abstract data types are the same as classes
 - B. Abstract data types do not allow inheritance
 - C. Classes cannot inherit from the same base class
 - D. Object have state and behavior
11. **SRS is also known as specification of:**
 - A. White box testing
 - B. Integrated testing
 - C. Stress testing
 - D. Black box testing
12. **A COCOMO model is:**
 - A. Common Cost Estimation Model
 - B. Complete Cost Estimation Model
 - C. Constructive Cost Estimation Model
 - D. Comprehensive Cost Estimation Model

13. Alpha and Beta Testing are forms of:
 - A. Acceptance testing
 - B. System testing
 - C. Integration testing
 - D. Unit testing
14. The tools that support different stages of software development life cycle are called...
 - A. CASE tools B. CAQE tools
 - C. CAME tools D. CARE tools
15. In an ER diagram an entity set is represent by a....
 - A. Rectangle B. Diamond box
 - C. Ellipse D. Circle
16. A report generator is used to:
 - A. Update files
 - B. Date entry
 - C. Print files on paper
 - D. Delete files
17. A form can be used to:
 - A. Modify records
 - B. Format printed output
 - C. Delete records
 - D. All of the above
18. The data dictionary tells the DBMS.
 - A. What files are in the database
 - B. What these files contain
 - C. What attributes are possessed by the data
 - D. All of the above
19. A file consists of:
 - A. Databases and fields
 - B. Records and sorts
 - C. Fields and sorts
 - D. Fields and records
20. Which of the following is record based logical model?
 - A. Network model
 - B. E-R model
 - C. Object oriented model
 - D. None of the above
21. A form of code that uses more than one process and processor, possibly of different type, and that may on occasions have more than one process or processor active at the same time, is known as:
 - A. Multiprogramming
 - B. Broadcasting
 - C. Multi-threading
 - D. Time sharing
22. A Memory:
 - A. Is a device that performs a sequence of operations specified by instructions in memory
 - B. Is the device where information is stored?
 - C. Is a sequence of instructions?
 - D. Is typically characterized by interactive processing and time of the CPU's time to allow quick response to each user
23. Banker's algorithm for resource allocation deals with.
 - A. Deadlock prevention
 - B. Deadlock recovery
 - C. Deadlock avoidance
 - D. Mutual exclusion
24. An instruction in a programming language that is replaced by a sequence of instructions prior to assembly or compiling is known as:
 - A. Procedure name
 - B. Micro
 - C. Label
 - D. Literal
25. Poor response times are caused by:
 - A. Processor busy
 - B. High paging rates
 - C. High I/O rate
 - D. Any of the above

26. **Capacity planning.**
- A. Requires detailed system performance information
 - B. Is independent of the operating system
 - C. Does not depend on the monitoring tools available
 - D. Is not needed in small installations
27. **The regular expression $(a|b)^*$ denotes the set of all strings.**
- A. With zero or more instances of a or b
 - B. Equal to regular expression $(a^*b^*)^*$
 - C. With one or more instances of a or b
 - D. Both A and C
28. **An automation is a... device and grammar is a... device.**
- A. Generative, cognitive
 - B. Acceptor, cognitive
 - C. Generative, acceptor
 - D. Cognitive, generative
29. **Which of the following problem is solvable?**
- A. Writing a universal Turing machine
 - B. Determining of an arbitrary Turing machine can be written for fewer than k instructions for some k
 - C. Determining of a universal Turing machine and some input will halt
 - D. All of the above
30. **If there exists a language L, for which there exists a TM, T, that accepts every word in L and either rejects or loops for every word that is not in L, is called...**
- A. Recursive
 - B. NP-HARD
 - C. Recursively enumerable
 - D. None of the above
31. **In an absolute loading scheme, which loader function is accomplished by programmer?**
- A. Allocation
 - B. Both A & B
 - C. Linking
 - D. Reallocation
32. **A compiler program written in a high-level language is called...**
- A. Source program
 - B. Machine language program
 - C. Object program
 - D. None of the above
33. **Advantages of using assembly language rather than machine language is that:**
- A. It is mnemonic and easy to read
 - B. Address any symbolic not absolute
 - C. Introduction to data to program is easier
 - D. All of the above
34. **The identification of common sub-expression and replacement of run-time computations by compile-time computations is:**
- A. Local optimization
 - B. Constant folding
 - C. Loop optimization
 - D. Data flow analysis
35. **In which of the following no information hiding is done?**
- A. Compiler prog 1, prog 2
 - B. Load R1, A
 - C. Run test, prog
 - D. 00100 1000010101
36. **Pixel phasing is a technique for:**
- A. Shading
 - B. Hidden line removal
 - C. Anti-aliasing
 - D. None of the above

37. Back face removal is an example of:
 A. Object space method
 B. Combination of both
 C. Image space method
 D. None of the above
38. Hue of a color is related to its:
 A. Luminance B. Incandescence
 C. Saturation D. Wave length
39. The wire is stretched to double its length, then its resistance R will become:
 A. $R/2$ B. $2R$
 C. $4R$ D. $R/4$
40. A diode can be used as a frequency multiplier because of its:
 A. Junction capacitance
 B. Avalanche voltage
 C. Non-linearity
 D. Forward breaks over
41. The combined resistance of two equal resistors connected in parallel is equal to:
 A. One half the resistance of original resistor
 B. Twice the resistance of one resistor
 C. Four times the resistance of one resistor
 D. One forth the resistance of one resistor
42. Which of the following is not characteristic of an oscillator?
 A. Negative feedback
 B. Reasonably high transistor gain
 C. Good output to input coupling
 D. Alternating current signal output
43. The color code of a resistor 2.7 K ohm with the tolerance of 10% is:
 A. Red, violet, red and silver
 B. Red, violet, orange, silver
 C. Red, violet, yellow and gold
 D. Red, violet, red and gold
44. The VA rating of a transformer is an expression of:
 A. The maximum frequency at which it can function
 B. The type of core material it has
 C. The voltage step-up or step-down ratio
 D. None of the above
45. A disadvantage of a half wave rectifier is the fact that:
 A. The output voltage is excessive compared to that of a full-wave rectifier
 B. The output current is excessive compared to that of a full-wave rectifier
 C. The output waveform is harder to filter than is the case with a full wave rectifier
 D. It requires several expensive diodes, whereas a full wave rectifier requires only a single node
46. AM and FM are example of...conversion.
 A. Digital-to-digital
 B. Analog-to-analog
 C. Digital-to-analog
 D. Analog-to-digital
47. In QAM, both.... Of a carrier frequency are varied.
 A. Frequency and amplitude
 B. Amplitude and phase
 C. Phase and frequency
 D. None of the above
48. If the baud rate is 400 for a QPSK signal, the bit rate is.... bps.
 A. 100 B. 800
 C. 400 D. 1600

49. If the bit rate for an ASK signal is 1200 bps, the baud rate is....

- A. 300 B. 600
- C. 400 D. 1200

50. Given an AM radio signal is 1200 bps, the baud rate is....

- A. 700 KHz
- B. 710 KHz
- C. 705 KHz
- D. Cannot be determined from given information

51. What is not an area of study research within Human Factors?

- A. Human anatomy
- B. Technology design
- C. Human performance
- D. Human and technology interaction

52. Google was found by:

- A. Larry Page and Sergey Brin
- B. Jeff Bezos
- C. David Filo and Jerry Yang
- D. Shawn Fanning

53. Archy is:

- A. Keyboard driven UI
- B. Mouse driven UI
- C. Both (A) & (B)
- D. Joystick driven UI

[HINT: Archy is a software system whose user interface introduced a different approach for interacting with computers with respect to traditional graphical user interfaces]

54. Which is the earliest social networking site?

- A. Classmates.com
- B. Facebook.com
- C. Orkut.com
- D. None of the above

55. User Interfaces are commonly programmed using.

- A. Application driven architecture
- B. Event driven architecture
- C. Network driven architecture
- D. None of the above

Answer question on the basis of the information below:

At a small company, parking spaces are reserved for the top executive: CEO, president, vice president, secretary and treasurer-with the spaces lined up in that order. The parking lot guard can tell at a glance if the cars are parked correctly by looking at the color of the cars. The cars are yellow, green, purple, red and blue and the executives' names are Alice, Bert, Cheryl, David and Enid.

- The car in the first space is red.
- A blue car is parked between the red car and the green car.
- The car in the last space is purple.
- The secretary drives a yellow car.
- Alice's car is parked next to David's.
- Enid drives a green car.
- Bert's car is parked between Cheryl's and Enid's.
- David's car is parked in the last space.

56. What color is the vice president's car?

- A. Green B. Blue
- C. Yellow D. Purple

57. Who is the CEO?

- A. Alice B. Cheryl
- C. Bert D. David

58. Who is the secretary?

- A. Enid B. Cheryl
C. David D. Alice

Five cities all got more rain than usual this year. The five cities are: Last Stand, Mile City, New Town, Olliopolis and Polberg. The cities are located in five different areas of the country: the mountains, the forest, the coast, the desert, and in a valley. The rainfall amounts were: 12 inches, 27 inches, 32 inches, 44 inches and 65 inches.

- The city in the desert got the least rain; the city in the forest got the most rain.
- New Town is in the mountains.
- Last Stand got more rain than Polberg, but less rain than New Town.
- Olliopolis got 44 inches of rain.
- The city in the mountains got 32 inches of rain; the city on the coast got 27 inches of rain.

59. Which city is in the desert?

- A. Last Stand B. New Town
C. Mile City D. Polberg

60. Which city got the most rain?

- A. Last Stand B. New Town
C. Mile City D. Olliopolis

61. How much rain did Mile City get?

- A. 12 inches B. 32 inches
C. 27 inches D. 44 inches

62. Where is Olliopolis located?

- A. The mountains B. In a valley
C. The coast D. The desert

Five roommates-Randy, Sally, Terry, Uma and Vernon-each do one housekeeping task-mopping, sweeping,

laundry, vacuuming or dusting-one day a week, Monday through Friday.

- Vernon does not vacuum and does not do his task on Tuesday.
- Sally does the dusting and does not do it on Monday or Friday.
- The laundry is done on Friday and not by Uma.
- Randy does his task on Monday
- Terry does his task, which is not vacuuming on Wednesday.

63. When does Sally do the dusting?

- A. Friday B. Tuesday
C. Monday D. Wednesday

64. What task does Terry do on Wednesday?

- A. Vacuuming B. Mopping
C. Dusting D. Sweeping

65. What day is the vacuuming done?

- A. Friday B. Tuesday
C. Monday D. Wednesday

66. Look at this series: 7, 10, 8, 11, 9, 12....., what number should come next?

- A. 7 B. 12
C. 10 D. 13

67. Fill in the blank in the middle of the series.

SCD, TEF, UGH....., WKL

- A. CMN B. VIJ
C. UJI D. IJT

68. The sum of ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?

- A. 4 years B. 10 years
C. 8 years D. None of above

69. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:
 A. $\frac{1}{4}$ B. $\frac{7}{15}$
 C. $\frac{1}{10}$ D. $\frac{8}{15}$
70. How much time will it take for an amount of Rs. 450 to yield Rs. 81 as interest at 4.5% per annum of simple interest?
 A. 35 yrs. B. 4.5 yrs.
 C. 4 yrs. D. 5 yrs.
71. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?
 A. 3.6 B. 8.4
 C. 7.2 D. 10
72. A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%?
 A. 3 B. 5
 C. 4 D. 6
73. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?
 A. 144° B. 168°
 C. 150° D. 180°
74. A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, then how old is B?
 A. 7 B. 9
 C. 8 D. 10
75. Which of the following is a prime number?
 A. 33 B. 93
 C. 81 D. 97
76. Which of the following divides the high-speed signal into frequency bands?
 A. T-switch
 B. Frequency division multiplexer
 C. Modem
 D. Time division multiplexer
77. A distributed network configuration in which all data/information pass through a central computer is:
 A. Bus network B. Duplex
 C. Star network D. Multiplex
78. Which layer of international standard organization OSI model is responsible for creating and recognizing frame boundaries?
 A. Physical layer B. Transport layer
 C. Data link layer D. Network layer
79. Which of the following device copies electrical signals from one Ethernet to another?
 A. Bridge B. Router
 C. Repeater D. Gateway
80. The X.25 standard covers how many OSI layers?
 A. 3 B. 7
 C. 4 D. None of above
81. How many bits internet address is assigned to each host on a TCP/IP internet which is used in all communication with the host?
 A. 16-bits B. 48-bits
 C. 32-bits D. 64-bits
82. Ability to take many forms is called....
 A. Polymorphism
 B. Member function
 C. Encapsulation
 D. Inheritance

83. A.... pointer can point to.... object:
 A. Derived class, base class
 B. Void, NULL
 C. Void, derived class
 D. Base class, derived class
84. For proper display of polymorphism, a method in the base class must be declared....
 A. Private B. Public
 C. Virtual D. All of the above
85. Which of these keywords is access specifies?
 A. Private and public
 B. Opened and closed
 C. Near and far
 D. Table and row
86. A suitable place to store class declarations is....
 A. Their own header file
 B. Auxiliary.cpp file
 C. Main.cpp files, along with function definitions
 D. None of the above
87. C programming is:
 A. Un structure programming
 B. OOPs
 C. Structure programming
 D. Web based programming
88. What are taken into account of state-space search?
 A. Postconditions B. Effects
 C. Preconditions D. Both B and C
89. Where does the Bayes rule can be used?
 A. Solving queries
 B. Decreasing complexity
 C. Increasing complexity
 D. Answering probabilistic query
90. What will take place as the agent observes its interactions with the world?
 A. Learning B. Perceiving
 C. Hearing D. Speech
91. Consider a linked list of n elements. What is the time taken to insert an element after an element pointed by some pointer?
 A. $O(1)$ B. $O(n)$
 C. $O(\log_2 n)$ D. $O(n \log_2 n)$
92. Consider that " n " element are to be sorted. What is the worst-case time complexity of bubble sort?
 A. $O(1)$ B. $O(n)$
 C. $O(\log_2 n)$ D. $O(n^2)$
93. Which of the following sorting procedures is the slowest?
 A. Quick sort B. Shell sort
 C. Heap sort D. Bubble sort
94. A full binary tree with n leaves contains:
 A. N nodes B. $2n-1$ nodes
 C. $\log_2 n$ nodes D. 2^n nodes
95. What is the number of nodes in a complete binary tree of level 5?
 A. 15 B. 63
 C. 25 D. 71
96. The average successful search time for sequential search on n items is:
 A. $n/2$ B. $(n+2)/2$
 C. $(n-1)/2$ D. $\log(n)+1$
97. The load instruction is mostly used to designate a transfer from memory to a processor register known as.....
 A. Accumulator
 B. Program counter
 C. Instruction register
 D. Memory address register

98. A group of bits that tell the computer to perform a specific operation is known as....

- A. Instruction code
- B. Accumulator
- C. Micro-operation
- D. Register

99. The time interval between adjacent bits is called the...

- A. Word-time
- B. Turnaround time
- C. Bit-time
- D. Slice time

100. A k-bit field can specify any one of...

- A. 3k registers
- B. K^2 registers
- C. 2k registers
- D. K^3 registers

ANSWER SHEET

1.A	2.A	3.C	4.C	5.A	6.B	7.C	8.C	9.D	10.B
11.D	12.C	13.A	14.A	15.A	16.C	17.D	18.D	19.D	20.A
21.C	22.B	23.C	24.B	25.D	26.B	27.A	28.D	29.A	30.C
31.D	32.A	33.D	34.B	35.D	36.C	37.A	38.D	39.C	40.C
41.A	42.A	43.D	44.D	45.C	46.A	47.B	48.B	49.D	50.A
51.B	52.A	53.A	54.A	55.A	56.A	57.B	58.D	59.D	60.A
61.C	62.B	63.B	64.D	65.C	66.C	67.B	68.A	69.D	70.C
71.C	72.B	73.D	74.D	75.D	76.B	77.C	78.C	79.C	80.A
81.C	82.A	83.D	84.C	85.A	86.A	87.C	88.D	89.D	90.A
91.A	92.D	93.D	94.B	95.B	96.B	97.A	98.A	99.C	100.C